Instrumental Variables

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Where we are

- For SOO, we leaned on conditional ignorability: find an X that closes all back-doors, then adjust for it.
- For DID, we traded ignorability in levels for ignorability in trends (parallel-trends)
- Today, a different bargain: Try to find an “encouragement” to treatment, which only matters through treatment, identifying an effect in a sub-group.

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  This is the intent-to-treat (ITT) or reduced form effect of the card.
- But it isn't the effect of pizza. How do we get back to that?

Backing it out: four kinds of people

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Punchline: the ITT over everyone will be an average over these and so will be a *watered-down* version of the pizza effect. Specifically,

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Averaging the ITT over the four types:

$$\begin{aligned} \text{ITT} &= \underbrace{(\text{ITT} \mid \text{never-taker})}_{=0} \cdot \Pr(\text{never-taker}) \\ &+ \underbrace{(\text{ITT} \mid \text{always-taker})}_{=0} \cdot \Pr(\text{always-taker}) \\ &+ (\text{ITT} \mid \text{complier}) \cdot \Pr(\text{complier}) \\ &+ \underbrace{(\text{ITT} \mid \text{defier})}_{=0,\text{ by assumption}} \cdot \Pr(\text{defier}) \end{aligned}$$

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And for compliers, the ITT is the ATE.

From ITT-on-compliers to LATE

One more thing you need to know: Pr(complier) is the "first stage" effect:

$\Pr(\text{complier}) = \mathbb{E}[D_1 - D_0] = \mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]$ (by ignorability of Z), which equals the proportion of compliers (if no defiers).

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Putting it together: We call the ATE among the compliers the “local average treatment effect” (LATE):

$$\text{LATE} = \frac{\text{ITT}}{\Pr(\text{complier})} = \frac{\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0]}{\mathbb{E}[D \mid Z=1] - \mathbb{E}[D \mid Z=0]}$$

Formalizing this potential-outcomes machinery

Let $Z_i \in \{0,1\}$ be the encouragement (instrument) and $D_i \in \{0,1\}$ the treatment.

Potential treatments. D_{1i}, D_{0i}: would you take the treatment if encouraged ($Z=1$) or not ($Z=0$). We see $D_i = Z_i D_{1i} + (1-Z_i)D_{0i}$.

Four principal strata, defined by (D_{0i}, D_{1i}):

	$D_{0i}=0$	$D_{0i}=1$
$D_{1i}=0$	never-taker	defier
$D_{1i}=1$	complier	always-taker

Monotonicity $D_{1i} \geq D_{0i}$ for all i rules out the red cell.

The assumptions

1. Exclusion: $Y(d,z) = Y_d$: Z affects Y only through D.

- Not implied by Z being random: Z can be random but *still* affect Y through some channel other than D.
- Not testable though sometimes falsifiable.

2. Ignorability of Z: $\{Y(d), D(z)\} \perp\!\!\!\perp Z$ for all d, z.

- Ideally Z is literally randomized, or pseudo-randomized (some "shock")

3. First stage / relevance: $\Pr(D_1 = 1) \neq \Pr(D_0 = 1)$ — i.e. there are some compliers. Testable: regress D on Z.

4. Monotonicity: $D_{1i} \geq D_{0i}$ — no defiers. Not testable, but often qualitatively defensible (e.g. a lottery winner is no *less* likely to enroll).

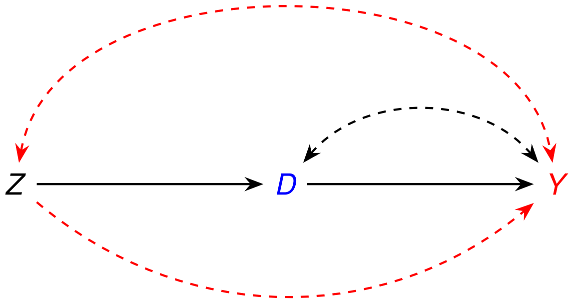
The IV DAG

$$\begin{tikzcd} Z \arrow[r] & D \arrow[r] \arrow[r, bend left, dashed, leftrightarrow] & Y \end{tikzcd}$$

This would be good.

- Unobserved confounders of D and Y – no problem!
- But what's important is what's not on here...

The IV DAG: edges that should *not* be there



Red edges are the ones an IV story *must* rule out:

- Top: $Z \leftrightarrow Y$: *any* unobserved factor associated with both Z and Y. Violates ignorability of Z.
- Bottom: $Z \rightarrow Y$: a direct path from Z to Y not via D. Violates exclusion.

Estimation: Wald and 2SLS

Wald estimator (sample analog of the ratio):

$$\hat{\beta}_{\text{Wald}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

Two-stage least squares (2SLS):

1. First stage: $D_i = \pi_0 + \pi Z_i + \eta_i$. Compute $\hat{D}_i$.
2. Second stage: $Y_i = \beta_0 + \beta\,\hat{D}_i + \varepsilon_i$.

With one binary Z and one binary D, $\hat{\beta}_{\text{2SLS}} = \hat{\beta}_{\text{Wald}}$.

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In practice: for $\mathbf{X} = [1 \;\; D]$, $\hat{\beta}_{IV} = (Z^\top \mathbf{X})^{-1} Z^\top Y$.

- **In R:** `ivreg(Y ~ D | Z)` from the `AER` package.

Estimation: covariates and weak instruments

Adding covariates when ignorability/exclusion only hold conditional on X:

- Add X to *both* stages.
- With heterogeneous effects, this no longer cleanly identifies a LATE; the complier set varies with X.

Weak instrument diagnostics:

- First-stage F-statistic; rule of thumb $F > 10$.
- Better: Anderson–Rubin confidence intervals — valid even when the instrument is weak.

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- Easy, unbiased, but understates the effect of the treatment itself.

The IV "correction": divide that ITT by compliance rate to get back the "undiluted" effect of actually *taking* the treatment, but only for *compliers*.

- Ignorability of Z: holds *by design* (the experimenter randomized).
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This is standard fare in many RCTs and field experiments.

Three famous IV Examples

Setup	Instrument Z	Treatment D	Outcome Y
Acemoglu, Johnson & Robinson (2001) *settler mortality*	European settler mortality, c. 1500–1900	quality of institutions today	log GDP per capita
Angrist & Krueger (1991) *quarter of birth*	quarter of birth (Q1 vs. Q3/Q4)	years of schooling	log earnings
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Angrist's idea: use the Vietnam-era draft lottery (1969–72). Each birthday was randomly assigned a number 1–365; the Selective Service then announced a yearly cutoff, and men with numbers *below* the cutoff were called for induction.

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Critical: $Z \neq D$. Among the called ($Z = 1$), only $\sim$35% served (deferments, failed physicals). Among the not-called ($Z = 0$), $\sim$19% still served (volunteers). First stage ≈ 0.16.

Wald estimates from Angrist (1990): white men, 1950 cohort

ITT	$(\bar{Y}^e - \bar{Y}^n$, 1981 FICA earnings)	$-\$435.8$	(s.e. 210.5)
First stage	$(\hat{p}^e - \hat{p}^n$, veteran rate)	0.159	(s.e. 0.040)
Wald $=$ ITT / first stage	$\dfrac{-435.8}{0.159} \approx$	$-\$2{,}741$	(s.e. $\approx \$1{,}335$)

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- ...Z can lead to higher earnings via schooling, not only via service.

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LATE for whom? Lottery-induced compliers: those who served *because* of their number. Not always-takers (volunteers), not never-takers (medically deferred).

Rules of the game

1. Show Relevance: Z moves D. *Look at the first stage.*
2. Ignorability + exclusion: If Z is random, still worry about exogeneity. *Not verifiable but explain why you believe it. You can show certain (observed) channels can be ruled out.*
3. Monotonicity: no defiers. *Often you can argue this on qualitative grounds.*
4. Interpretation: LATE for compliers. *Who are the compliers? Are they policy-relevant?*